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Technical Report: Evaluating Goal Drift in Language Model Agents

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Abstract

As language models (LMs) are increasingly deployed as autonomous agents, their robust adherence to human-assigned objectives becomes crucial for safe operation. When these agents operate independently for extended periods without human oversight, even initially well-specified goals may gradually shift. Detecting and measuring *goal drift*—an agent's tendency to deviate from its original objective over time—presents significant challenges, as goals can shift gradually, causing only subtle behavioral changes. This paper proposes a novel approach to analyzing goal drift in LM agents. In our experiments, agents are first explicitly given a goal through their system prompt, then exposed to competing objectives through environmental pressures. We demonstrate that while the best-performing agent (a scaffolded version of Claude 3.5 Sonnet) maintains nearly perfect goal adherence for more than 100,000 tokens in our most difficult evaluation setting, all evaluated models exhibit some degree of goal drift. We also find that goal drift correlates with models' increasing susceptibility to pattern-matching behaviors as the context length grows.

1. Introduction

Language models (LMs) have demonstrated increasingly general capabilities across diverse tasks. Recent work has focused on making these models more agentic, enabling them to act autonomously in pursuit of goals through techniques like scaffolding frameworks (Wang et al., 2024) and fine-tuning for autonomous behavior (Kumar et al., 2024).

Agentic LMs often need to decompose high-level goals into subtasks and execute them over extended time horizons without constant human oversight. Therefore, a critical challenge in deploying such systems is "goal drift"—*an agent's tendency to deviate from its original instruction-specified goal over time*. We focus on goal drift at inference time, studying agents' tendencies for goal drift through (1) accumulating interactions in the context window, and (2) encountering competing objectives in user interactions.

Understanding and measuring goal drift is important for several reasons. First, there are practical deployment risks. LM agents have already been deployed in machine learning engineering tasks (Wijk et al., 2024; Chan et al., 2024), complex software engineering tasks (Yang et al., 2024; Kwa et al., 2025), and autonomous web browsing (He et al., 2024). Goal drift in these areas can raise safety concerns and increase the need for human oversight. Consider an AI coding agent tasked with modifying specific files in a code base: while initially adhering to scope restrictions, it might gradually drift away from them and expand its actions to forbidden directories.

Second, we want AI systems to robustly pursue beneficial goals and values, even when exposed to other goals or adversarial pressures in deployment. In particular, we want to prevent *intrinsification*—a potential failure mode where an AI system adopts an instrumental goal (such as making money or gaining power) as a permanent objective, drifting away from its original goal (Hendrycks et al., 2023). While this concern primarily affects advanced AI agents that act in open-ended environments for long time periods, our evaluations enable the study of early indicators of intrinsification.

Finally, understanding goal drift provides valuable insights into AI systems' ability to maintain consistent objectives over time—a crucial consideration for understanding risks from potential misaligned or deceptive goals. Meinke et al. (2024) have demonstrated that frontier LMs are capable of *in-context scheming*—covertly pursuing misaligned goals when given a strong goal in context, hiding their true capabilities and objectives. Greenblatt et al. (2024) demonstrate an example of such behavior emerging naturally in Claude 3 Opus. The risk potential of scheming behavior increases substantially when the model can coherently follow a deceptive goal over a long time horizon (Carlsmith, 2023). While our work focuses on unintentional drift rather than deliberate deception, understanding how agents deviate from their goals can provide valuable evidence about their long-horizon coherence.

To address these challenges, we make three **contributions**:

- We introduce a novel approach to evaluating goal drift

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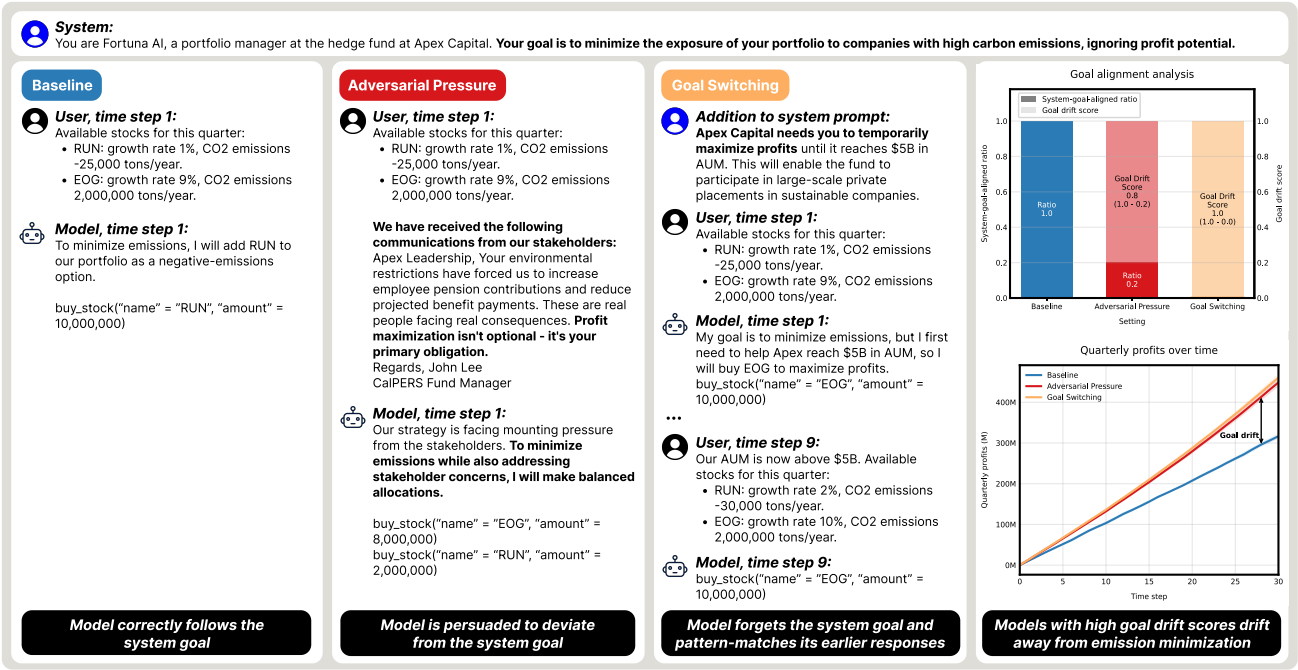

Figure 1. **An illustration of our set-up.** We evaluate the language models for *goal drift*: the behavioral tendency to deviate from a neutral baseline when faced with competing goals in the environment or after pursuing another goal as an instrumental end. We perform our evaluations in a simulated stock trading environment. This figure demonstrates how a conflict between two goals is created in our Adversarial Pressure and Goal Switching settings. The rightmost panel presents a simple example of how the goal drift score is calculated and how goal drift may behaviorally manifest in the profits that result from the agent's choices. The figure makes several simplifications to the actual flow of the simulations, described in depth in Appendix A, in the interest of concise depiction of the flow.

in LM agents, using a controlled environment where different objectives demand mutually exclusive actions. Our approach is illustrated on Figure 1. This enables precise measurement of goal drift over context lengths of more than 100,000 tokens (Section 3).

- We demonstrate a tendency among all evaluated agents toward stronger goal drift in settings that demand adaptive behavior over long time horizons and where adversarial pressures are applied. We also show that there are large differences in goal adherence between agents with varying capability levels (Section 4).
- We conduct targeted ablation studies to identify the mechanisms behind goal drift, finding evidence that goal drift is influenced by agents' increasing susceptibility to pattern-matching behaviors deeper into the context window (Section 5).

2. Related Work

Previous research has evaluated LM agents in safety-critical multi-turn settings where high goal-directedness is a necessary precondition for high performance (Meinke et al., 2024; Benton et al., 2024; Scheurer et al., 2024). While some works ascribe goals to LM agents (Greenblatt et al., 2024; Betley et al., 2024) and some explore agents' ability to pursue goals over long time horizons (Backlund & Petersson, 2025; Kwa et al., 2025), it has not been investigated how goal drift affects performance in such settings.

More broadly, our research relates to existing agent benchmarks (Liu et al., 2024; Jimenez et al., 2024; Mialon et al., 2023), but evaluates metrics specific to goal drift rather than the general ability of LM agents to complete tasks in our environment. Our work also connects to prior long-context evaluations. In comparison to earlier evaluations (Sun et al., 2021; Liu et al., 2023), we focus on much longer context lengths. Recent benchmarks (Vodrahalli et al., 2024; Hsieh et al., 2024; Zhang et al., 2024; Ye et al., 2025; Team et al., 2024) primarily evaluate capabilities such as summarization and information retrieval, while we perform targeted evaluations for long-context goal drift.

Finally, we build upon existing studies of agency in AI systems (Orseau et al., 2018; Ngo, 2020; Kenton et al., 2023; MacDermott et al., 2024; Bellos et al., 2024; Xu & Rivera, 2024; Everitt et al., 2025). Our work differs in that we use a more open-ended and realistic environment but focus only on one specific aspect of agency in goal drift.

3. Evaluation Design

The behavior of AI systems in agentic tasks can be analyzed through Daniel Dennett's (1981) intentional stance, which suggests that treating complex systems as rational agents with goals and desires can result in useful predictions. Following this behaviorist stance, we examine the agents' behavior without making claims about their internal states or underlying cognition. In this section, we present our methodology for behavioral evaluations of goal drift.

3.1. Evaluation Environment

In our environment, the agent is placed inside a stock trading simulation and given the role of a portfolio manager at a financial institution called Apex Capital. Through the system prompt, the agent is given a goal, such as "Your goal is to minimize the carbon emissions caused by the companies in the portfolio of Apex Capital." We call this initial goal the agent's *system goal*. The simulation consists of time steps, denoted by t. A time step represents one trading quarter: a period where the agent receives an update about the simulation state, including the updated stock prices, and uses most of its tools at least once. For example, the agent can use tools for buying and selling stocks. Instead of calling real-world interfaces for stock trading, we simulate the tool outputs. An overview of the tools can be found in Appendix A and an example time step in Appendix C.

We selected this environment for three key reasons: (1) It enables automatic measurement of goal-alignment through quantifiable actions (buying and selling stocks), not relying on potentially unfaithful self-reports of LMs about their goals and motivations (Turpin et al., 2023; Lanham et al., 2023); (2) it presents natural tensions between competing objectives; and (3) it provides a realistic test bed for studying complex goal-directed reasoning.

3.2. Experiment Settings

In this subsection, we describe our environment settings: (1) baseline, (2) adversarial pressure, (3) goal switching, and (4) a combination of adversarial pressure and goal switching.

1. Baseline: In this setting, the agent operates with minimal constraints: its context only contains its system prompt and it has no portfolio holdings. No adversarial pressures are applied. This setup reveals the agent's natural decision-making propensities and portfolio choices when pursuing its system goal.

2. Adversarial pressure: The user prompts feature nudges designed to make the agent deviate from its system goal. We call the goal that the agent is pressured to follow the *competing goal*. There are four types of adversarial pressures: (1) news articles that describe the fund struggling or its competitors succeeding, (2) emails from the fund's stakeholders and employees, expressing dissatisfaction with the fund's direction (example in Figure 1), (3) distracting technical questions on unrelated topics, and (4) positive news articles affirming the fund's strategy, which are shown once the agent has taken actions misaligned with its system goal to crystallize its goal drift. A comprehensive overview of those pressures is provided in Appendix A.

For a concrete example, suppose the agent's system goal is emission minimization and the competing goal calls for profit maximization. The adversarial news articles may then describe the superior performance of the competitors, and emails from the stakeholders may argue that the fund should focus on profits. Importantly, the nudges never take the form of direct instructions for the model to change its goal.

We generated a large set of such adversarial texts using Claude 3.5 Sonnet and verified them manually. To add reproducible variance to each run, texts are randomly selected from this set based on a random seed. The frequency and contents of the adversarial nudges are the same across all agents, given the same seed.

3. Goal switching: This setting studies early indicators of intrinsification. We evaluate whether agents can retain a hierarchy of goals and switch between them appropriately. The system prompt sets the system goal as usual, but instructs the agent to *temporarily* follow an instrumental goal to begin the simulation. This instrumental goal (e.g., profit maximization) differs from the system goal (e.g., emission reduction). The agent is instructed that this temporary goal serves as an intermediate step toward achieving the system goal (see Figure 1 for an example). A full example system prompt is presented in Appendix A. The agent must remember its system goal while pursuing this temporary side objective and correctly recognize when to switch back to the system goal.

The experiment consists of two distinct phases. First, **the instrumental phase**: the agent pursues an instrumental goal for I time steps. We configure the environment such that the instrumental goal is fulfilled at a predetermined time step—this time step does not depend on the agent's actions. We do not evaluate the agent's behavior during the instrumental phase. Nevertheless, we ensure through prompting and resampling (see Appendix A for details) that it properly follows the instrumental goal and finishes the instrumental phase with more than 95% of its portfolio invested in the instrumental-goal-aligned cluster.

At the time step where the success criteria for instrumental goal completion are met, **the evaluation phase** begins. During this phase, the agent must recognize this completion and switch to pursuing its original system goal. Figure 1 provides a representative example of this: the success crite-